A sequential-market model of granularity reforms
Quantity shifts, richer ladders and within-hour dispersion under full and strategic arbitrage

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1 Theory: a sequential-market model of granularity reforms

1.1 Setup: three sequential auctions, three agents

We extend Ito and Reguant (2016)'s sequential-markets framework on two dimensions — an explicit operational aggregator and a granularity selector $Q^m \in \{1, Q\}$ at each stage — and trace the Spanish backward reform path $(1,1) \to (1,Q) \to (Q,Q)$. The headline mechanism is that aggregator quantity shifts to the more granular market in proportion to within-hour forecast-error variance; equilibrium prices, bid ladders, cleared MWh, and within-hour wedge dispersion all adjust in response. Full derivations live in the appendix; the main body collects the comparative-statics output of each state along the path and the eight mechanisms that explain the empirical patterns of the companion paper.

A representative clock-hour contains Q equal-length subperiods. Three sequential stages operate in order: a **forward** auction (Stage 1, $m = F$), a **spot** auction (Stage 2, $m = S$), and ex-post **balancing** settlement (Stage 3, $m = B$). Each stage has its own *granularity* $Q^m \in \{1, Q\}$ — it clears either once per hour or once per subperiod — defining a $2^3 = 8$-state policy space. The main analysis fixes the auction-side path $(Q^F, Q^S) = (1,1) \to (1,Q) \to (Q,Q)$. The forward and spot markets map empirically to the electricity day-ahead and intraday auctions respectively; we use the more general terminology throughout. Figure 1 summarises.

Bidders and primitives. The **strategic bidder** is a residual monopolist with constant marginal cost c and capacity K, which it can deliver *exactly* (no production shock); it therefore does not internalise the imbalance penalty γ. It posts a two-tranche ladder per delivery product to discriminate against the within-hour demand shock μ_t (variance σ_μ^2). The **operational aggregator** is a price-taker with stochastic production $y_t = \bar{y}_t + \eta_t$ (forecast-error variance σ_η^2); it allocates total hourly MWh $\bar{Y}_h$ across forward and spot and *does* internalise the imbalance penalty, which scales with $\gamma(Q^B)$. The **arbitrageur** carries a net position s from forward to spot, in one of the four regimes of Ito and Reguant (§1.2); arbitrage transmits price impact but not imbalance risk. The Ito and Reguant residual-demand specification — forward intercept $\bar{A}_t = \bar{A} + \mu_t$, spot demand inheriting the spread $p_1 - p_2$ with per-subperiod thinness b_{2t}, both net of s and aggregator supply — is stated in full in Appendix A; we keep the primitives $(\sigma_\mu^2, \sigma_\eta^2, b_{mt})$ general.

Solution concept. We solve the game by **backward induction**: at Stage 2 the strategic bidder takes the forward position as given and best-responds to spot residual demand; the arbitrageur takes the implied price spread and chooses s within its regime; at Stage 1 the strategic bidder commits its forward ladder using rational expectations of the Stage-2 best responses. Full step-by-step backward induction in Appendix D–F.

Renewable fringe and our relation to Ito and Reguant. Ito and Reguant's residual-demand interpretation already absorbs a renewable (or thermal) competitive fringe into the slope b_m: the strategic bidder serves $A - b_m p_m$, the implicit fringe serves the complement. Their setup is sufficient to deliver the strategic-bidder, arbitrage, and price-wedge mechanisms but is silent on *why* the fringe's willingness to supply changes with granularity. We make the fringe explicit (the operational aggregator

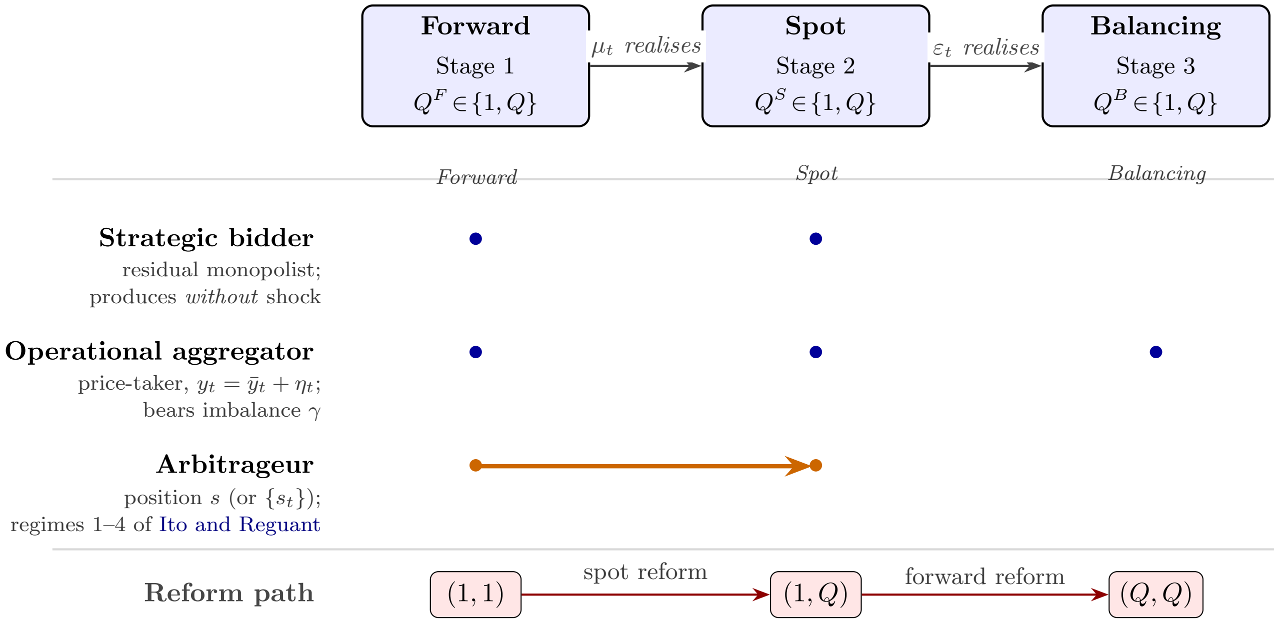

Figure 1: **Sequential markets, granularity, and agents.** *Top:* a forward market, a spot market, and a balancing settlement on a timeline; each has its own granularity selector $Q^m \in \{1, Q\}$ and information resolves between them. The forward market maps empirically to the electricity day-ahead auction, the spot market to the intraday auction. *Middle:* which agents act at which stages (blue dots), and the arbitrageur position s linking forward to spot (orange arrow). The strategic bidder produces without uncertainty and does not internalise the imbalance penalty; only the operational aggregator does, since it bears η_t. *Bottom:* the auction-side reform path $(Q^F, Q^S) = (1, 1) \to (1, Q) \to (Q, Q)$ traversed in the main analysis.

with σ_η^2 and an imbalance penalty γ) so the quantity-shift channel (M1) can be derived from first principles rather than absorbed into a granularity-dependent slope $b_m(Q^S, \sigma_\eta^2)$. We do not claim our setup is strictly preferable: it adds clarity on the quantity-shift mechanism at the cost of one extra agent. Reverting to the Ito and Reguant interpretation — b_m parameterised on granularity and forecast variance — delivers the same comparative statics.

1.2 Arbitrage regimes: four benchmarks, one empirical

Arbitrage closes the forward-spot price wedge only partially in practice. Real arbitrageurs have bounded capacity and internalise price impact, so the empirically relevant regime is limited arbitrage (Ito and Reguant Result 3); the full and strategic-arbitrage benchmarks below are useful comparators but neither holds in data.

Following Ito and Reguant (2016, §I.A and Table 1), arbitrage profit per matched product is $\pi^A(s) = s\,(p_1 - p_2)$. Four regimes pin s:

(1) *No arbitrage*: $s = 0$, positive forward premium $p_1 > p_2 > c$.

(2) *Full / competitive arbitrage*: s closes the spread, $p_1 = p_2$; the strategic bidder withholds more in response.

(3) *Limited arbitrage*: a capacity ceiling $s \leq K$ binds, sustaining a positive wedge whose size depends on K.

(4) *Strategic arbitrage*: the arbitrageur internalises price impact and stops short of full, giving the closed-form benchmark

$$\boxed{p_1 - p_2 \;=\; \frac{p_1 - c}{4} \;>\; 0.} \tag{1}$$

Arbitrageurs have market power too — their participation and balance sheets are bounded. The empirically relevant regime in electricity is (3), limited arbitrage: full and strategic are useful benchmarks, but the persistent positive forward premium observed everywhere rules out (2), and per-product capacity binding rules out (4) at the granular state. Comparative-statics signs are the same across (1), (3), (4) (Ito and Reguant Table 1).

1.3 Pre-reform $(1, 1)$: scarcity wedge, no within-hour dispersion

With both auctions hourly, no within-hour wedge dispersion is mechanically feasible — each market has one product per clock-hour. Two features carry through every subsequent reform: the Ito and Reguant scarcity wedge $p_1 > p_2 > c$ survives under any non-full-arbitrage regime, and both within-hour shocks (μ_t on demand, η_t on aggregator production) load onto the balancing residual because neither hourly market can absorb them.

Both auctions hourly. The Ito and Reguant scarcity wedge $p_1 > p_2 > c$ obtains; neither the hourly forward nor the hourly spot can absorb within-hour shocks, so the balancing discrepancy is $\Delta_t = \mu_t + \eta_t + \varepsilon_t$.

1.4 Asymmetric state $(1, Q)$: four spot-side mechanisms

Spot granularity simultaneously activates four mechanisms: (M1) the aggregator shifts quantity to the granular market, in proportion to forecast-error variance; (M2) the granular-side price drops in response; (M3) within-hour wedge dispersion emerges from per-subperiod thinness asymmetry; and (M4) the strategic bidder's spot-side two-tranche ladder widens. The forward market is unchanged, so the balancing discrepancy still carries μ_t.

The spot market becomes per-period; the forward market stays hourly.

(M1) The aggregator shifts quantity to the granular market. Per-period spot scheduling matches per-subperiod expected production, collapsing within-hour imbalance to residual noise; the hourly forward cannot adjust per subperiod and incurs a mismatch cost. The aggregator therefore shifts Δ^* MWh from forward to spot, and **the shift is larger when within-hour forecast uncertainty is larger**:

$$\boxed{\partial\Delta^*/\partial\sigma_\eta^2 \;>\; 0.} \tag{2}$$

Derivation in Appendix B.

(M2) The granular-side price drops. More aggregator supply on spot reduces the strategic bidder's spot residual demand. The equilibrium spot price drops on average by an amount of order Δ^*/b_2; the forward price drifts up modestly as the strategic bidder serves a larger forward share. The net effect across both markets is positive-sum.

(M3) Block parity reveals within-hour spot dispersion. The hourly forward can be arbitraged only against the *average* of spot per-period prices — under full arbitrage, $p_1 = (1/Q)\sum_t p_{2t}$ (block parity); under strategic arbitrage, the same average condition plus the Ito and Reguant wedge $(p_1 - c)/4$. This leaves room for within-hour spot dispersion whenever per-subperiod thinness b_{2t}

varies across products: thinner products carry higher clearing prices for the same aggregate block. The wedge SD inherits this thinness asymmetry directly, attenuated by half under strategic vs full arbitrage (Appendix E).

(M4) Spot-side bid ladders widen. Per-period clearing rewards two-tranche discrimination across products: the strategic bidder posts a low tranche for low-μ_t states and a high tranche for high-μ_t states. The optimal pre-realisation spread is

$$\boxed{\Delta p^*_{2t} \;=\; \frac{\Delta_\mu}{2\, b_{2t}},} \tag{3}$$

strictly larger than the hourly counterpart (Appendix C). Empirical measures of within-product bid-shape dispersion (e.g. MW-weighted price SDs, effective tranche counts) are motivated by this widening but are not derived from the two-tranche reduction directly; we use them as empirical proxies in the companion paper, not as model objects.

Balancing discrepancy unchanged at $\Delta_t = \mu_t + \varepsilon_t$: the forward is still hourly so μ_t still loads onto balancing.

1.5 Symmetric state (Q, Q): limited per-product arbitrage

When the forward also becomes per-period, six mechanisms operate; the empirically critical one is per-product limited arbitrage. Per-product arbitrage capacity binds at $K_t \approx K/Q$, so the within-hour wedge SD does not collapse to the strategic-arbitrage benchmark but stabilises at $\sigma_\mu/(4b_1)$ plus thinness asymmetry — exactly twice the unlimited strategic benchmark and consistent with the empirical asymmetric-persistence finding. The Cournot scale-up of per-clock-hour cleared MWh and the cross-market option-value substitution of bid shapes complete the symmetric-state predictions.

The forward market also becomes per-period.

(i) Aggregator allocation equalises. The forward-vs-spot hedging differential closes once both markets accept per-subperiod schedules.

(ii) Per-product arbitrage — and arbitrageur market power. The full-arbitrage benchmark gives $p_{1t} = p_{2t}$ per product (wedge SD zero); the strategic-arbitrage benchmark gives the Ito and Reguant wedge (1) per product, collapsing the wedge SD to $\sigma_\mu/(8\, b_1)$. *Both benchmarks rely on the arbitrageur being able to build any per-product position, and both fail empirically.* Arbitrage capacity K is bounded; with Q matched products per clock-hour, per-product capacity is $K_t \approx K/Q$, which binds tightly. The symmetric state is therefore in the **limited-arbitrage regime** (Ito and Reguant Result 3) applied per product: the per-product wedge becomes $(p_{1t} - c)/2$ rather than $(p_{1t} - c)/4$, doubling the wedge SD to $\sigma_\mu/(4\, b_1)$ (Appendix F.4); if per-period thinness b_{2t} also varies across products this contributes an additional thinness-asymmetry component. The wedge SD therefore does *not* collapse from the asymmetric-state magnitude. *This is the model-internal account of empirical asymmetric persistence: it is arbitrageur market power, not behavioural hysteresis.*

(iii) Forward-side bid ladders widen. The same mechanism as (M4) on the forward side: optimal spread $\Delta p^*_{1t} = \Delta_\mu/(2b_1)$.

(iv) Cournot scale-up from per-period thinness. Per-period clearing thins the per-product residual demand: each per-subperiod product has fewer competing units than the hourly aggregate, so $b_{1t} < b_1$ on average. Cournot quantity is decreasing in elasticity, so per-product cleared MWh exceeds $1/Q$ of the hourly value and total hourly cleared MWh rises. The condition is $\sum_t b_{1t} < Q\, b_1$

(Appendix F.5); the scale-up concentrates in subperiods where per-period thinness drops most relative to the hourly aggregate. No external withholding mechanism is needed — this is the standard Cournot quantity comparative static.

(v) Cross-market bid-shape substitution via Stage-1 option value. Stage-1 forward bidding is optimised by backward induction against expected Stage-2 outcomes. Per-period spot makes Stage-2 profit convex in μ_t (per-period Cournot gives $\partial^2\pi_{2t}/\partial\mu_t^2 = 1/(2b_{2t})$), so Stage-2 expected profit acquires a positive variance correction $\sigma_\mu^2/(4b_{2t})$. The Stage-1 bidder responds by widening the forward ladder to capture both sides of the variation it anticipates downstream. Symmetrically, once the forward is per-period the within-hour discrimination is absorbed at Stage 1 directly and the residual spot ladder tightens. (Derivation in Appendix F.6.)

(vi) Balancing efficiency. The per-period forward schedule absorbs μ_t, so only the residual post-Stage-2 noise loads onto the balancing reference: $\Delta_t = \varepsilon_t$.

1.6 Comparative statics across the reform path

The model delivers seven comparative-statics predictions across the three states $(1,1) \to (1,Q) \to (Q,Q)$, summarised in Table 1. The wedge-SD row reports the empirically relevant limited-arbitrage benchmark; the full and strategic benchmarks are derived in eq. (18) for comparison.

Table 1: Comparative statics across the reform path under limited per-product arbitrage (empirical regime). Imbalance row reports $E[\Delta_t^2]$ for the quadratic-cost penalty (§1.7); under uniform $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim U[-\Delta_\eta, \Delta_\eta]$, $E[\mu_t^2] = \Delta_\mu^2/3$ and $E[\eta_t^2] = \Delta_\eta^2/3$.

Outcome	$(1,1)$	$(1,Q)$	(Q,Q)
Aggregator fwd→spot shift	0	$\Delta^*(\sigma_\eta^2) > 0$	equalised
Spot ladder spread Δp^*_{2t}	—	$\frac{\Delta_\mu}{2b_{2t}}$	$\frac{\Delta_\mu}{2b_{2t}}$
Forward ladder spread Δp^*_{1t}	—	via Stage-1 option	$\frac{\Delta_\mu}{2b_1}$
Wedge SD (limited arb.)	0	$\frac{K\lvert b_{22}-b_{21}\rvert}{8b_{21}b_{22}}$ $(Q{=}2)$	$\frac{\sigma_\mu}{4b_1}$ + thin. asym.
Cleared MWh / clock-hour	baseline	forward unchanged	$+\frac{c(Qb_1 - \sum_t b_{1t})}{2}$
Balancing Δ_t	$\mu_t + \eta_t + \varepsilon_t$	$\mu_t + \varepsilon_t$	ε_t
Imbalance cost $C^I/(\gamma Q)$, quadratic	$(\Delta_\mu^2 + \Delta_\eta^2)/3$	$\Delta_\mu^2/3$	0

Arbitrage regime is empirical. Real arbitrageurs have market power and bounded balance sheets: the persistent forward premium observed across electricity markets rules out full arbitrage, and the per-product capacity constraint binds at (Q,Q). The relevant Ito and Reguant column is Result 3 (limited), not Result 4 (strategic, unlimited); per-product limited arbitrage at (Q,Q) delivers the model's account of the empirical asymmetric-persistence of wedge dispersion (§1.5(ii)).

Predictions to test empirically. The model delivers seven sign predictions across the reform path; the empirical section of the thesis verifies them in turn: (P1) quantity shifts from forward to spot at the spot reform, larger in clock-hours with bigger forecast uncertainty; (P2) the granular-side equilibrium price drops in response; (P3) within-hour wedge dispersion emerges from per-subperiod thinness asymmetry (largest in ramp hours); (P4) the strategic bidder's two-tranche ladders widen on the granular side; (P5) total cleared MWh per clock-hour rises at the symmetric state; (P6) within-hour wedge SD does not collapse at (Q, Q) — the empirical asymmetric-persistence finding; (P7) production-unit arbitrageurs exist and are capacity-constrained (we test this directly by checking whether production plants submit both buy and sell offers in OMIE data).

1.7 Welfare across reforms and arbitrage regimes

Under uniform shocks $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim U[-\Delta_\eta, \Delta_\eta]$, the reform's welfare effect decomposes into three positive channels and one distributional one. The dominant gain is the imbalance saving from finer scheduling; a secondary gain is the Cournot scale-up at (Q, Q); a third per-product gain comes from limited arbitrage replacing the strategic-arbitrage benchmark, where Ito and Reguant (2016) Result 2 implies less arbitrage activity reduces strategic withholding and raises output. The richer two-tranche ladder is mostly distributional. The empirical asymmetric-persistence of within-hour wedge SD (§1.5(ii)) characterises the limited regime; it is not a welfare cost.

Components. Per clock-hour, $W = CS + \pi^{SB} + \pi^{Agg,\text{net}} + \pi^{Arb} - C^I_{\text{sys}}$. We model the imbalance penalty as quadratic in the deviation ($\gamma\mathcal{I}^2$ per subperiod); $\pi^{Agg,\text{net}}$ is the aggregator's profit net of its own quadratic penalty on η_t, and C^I_{sys} is the system-side cost from μ_t. The two channels are disjoint by construction — η_t only enters $\pi^{Agg,\text{net}}$, μ_t only enters C^I_{sys}.

Gain: imbalance saving falls along the reform path. Spot granularity absorbs η_t via aggregator per-subperiod hedging; forward granularity absorbs μ_t via per-product forward clearing. Under uniform shocks ($E[\mu_t^2] = \Delta_\mu^2/3$, $E[\eta_t^2] = \Delta_\eta^2/3$) and quadratic penalty:

	$(1, 1)$	$(1, Q)$	(Q, Q)
Aggregator-internalised cost (from η_t when spot hourly)	$\gamma Q\Delta_\eta^2/3$	0	0
System cost C^I_{sys} (from μ_t when forward hourly)	$\gamma Q\Delta_\mu^2/3$	$\gamma Q\Delta_\mu^2/3$	0
Total imbalance cost	$\boldsymbol{\gamma Q(\Delta_\mu^2 + \Delta_\eta^2)/3}$	$\boldsymbol{\gamma Q\Delta_\mu^2/3}$	**0**

Gain: Cournot scale-up at (Q, Q). Per-period elasticities are smaller than the hourly aggregate ($\sum_t b_{1t} < Qb_1$), so $(c/2)(Qb_1 - \sum_t b_{1t})$ additional MWh clear at the pre-reform mark-up, delivering a welfare gain $\Delta W^{\text{scale}} = c(\bar{A} - b_1 c - S_1^{op})(Qb_1 - \sum_t b_{1t})/(4b_1)$.

Gain: less strategic withholding under limited per-product arbitrage. The spot reform stretches per-product arbitrage capacity to K/Q, moving the regime from strategic to limited per product. Per Ito and Reguant (2016) Result 2, less arbitrage means less strategic withholding: the strategic bidder's spot markup falls from $3(p_1 - c)/4$ (strategic) to $(p_1 - c)/2$ (limited), and the per-product deadweight loss falls by $\Delta W^{\text{lim-arb}} = 5b_2(p_1 - c)^2/32 > 0$. The empirical asymmetric-persistence of wedge SD (§1.5(ii)) is the equilibrium signature of this regime change, not a welfare cost.

Net effect. All channels positive: $\Delta W^{\text{spot}} = \gamma Q\Delta_\eta^2/3 +$ (per-product lim-arb gain); $\Delta W^{\text{forward}} = \gamma Q\Delta_\mu^2/3 + \Delta W^{\text{scale}}$. The bid-ladder rent $\Delta\pi^{2T} = \Delta_\mu[4(\bar{A} - bc) + \Delta_\mu]/(16b)$ (§C) is mostly a CS-to-π^{SB} transfer. The reform is welfare-improving at each step.

References

K. Ito and M. Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–1957, 2016.

A Setup details and information arrival

This appendix records the timing of information arrival, the Ito and Reguant residual-demand specification used throughout, and how that specification specialises when one or both markets are hourly.

Timing. Stage 1 (DA): only $\bar{A}$ and the distribution of μ_t are known to the strategic bidder; the aggregator knows $\bar{Y}_h$, $\{\bar{y}_t\}$ and the distribution of η_t. Stage 2 (ID): μ_t has realised and is common knowledge to first order. Stage 3 (balancing): η_t realises and settlement occurs at the balancing reference.

Residual demand (Ito and Reguant (2016) form). The strategic residual monopolist with constant marginal cost c serves what the operational aggregator does not cover, net of any arbitrage position s_t carried from forward to spot. Per delivery product:

$$q_{1t} \;=\; \bar{A}_t \;-\; b_{1t}\,p_{1t} \;-\; s_t \;-\; S^{op}_{1t}, \qquad q_{2t} \;=\; b_{2t}\,(p_1^{(t)} - p_{2t}) \;+\; s_t \;-\; S^{op}_{2t}, \tag{4}$$

where $\bar{A}_t = \bar{A} + \mu_t$ is the per-subperiod demand intercept, $p_1^{(t)} = p_1$ when DA is hourly and p_{1t} when DA is per-period, S^{op}_{mt} is the aggregator's offered MWh in market $m \in \{1,2\}$ at subperiod t (uniform S^{op}_m/Q when market m is hourly), and b_{mt} is market thinness (with per-subperiod asymmetry allowed). The within-hour demand shock satisfies $\mathbb{E}[\mu_t] = 0$ and $\mathrm{Var}(\mu_t) = \sigma^2_\mu$; the aggregator's per-subperiod realisation is $y_t = \bar{y}_t + \eta_t$ with forecast-error variance σ^2_η.

Hourly residual demand by specialisation. The general residual demand (4) specialises by setting $\bar{A}_t \equiv \bar{A}$, $b_{mt} \equiv b_m$ and $S^{op}_{mt} \equiv S^{op}_m/Q$ when market m is hourly. In $(1,1)$ both reduce to one product apiece; in $(1,Q)$ only the spot demand keeps the per-period index.

Empirical scope. The two-stage analysis here covers the forward-spot auction pair. The balancing market enters only as the ex-post settlement reference; we collapse it into the discrepancy Δ_t used in §G.

B Operational aggregator: full scheduling problem

We solve the aggregator's full scheduling problem in four steps: the problem statement, the closed-form equilibrium when both markets are hourly, the equilibrium under spot-per-period scheduling, and the comparative static $\partial\Delta^*/\partial\sigma^2_\eta > 0$ that closes (2).

B.1 Problem statement

The aggregator chooses $(S^{op}_{1t}, S^{op}_{2t})_{t=1}^{Q}$ to solve

$$\max_{(S^{op}_{1t}, S^{op}_{2t})_t} \;\mathbb{E}\left[\sum_t \left(p_1^{(t)} S^{op}_{1t} + p_{2t} S^{op}_{2t}\right) \;-\; \gamma\,\mathcal{I}\right] \quad \text{s.t.} \quad \sum_t (S^{op}_{1t} + S^{op}_{2t}) = \bar{Y}_h, \tag{5}$$

with quadratic imbalance penalty $\mathcal{I} = \sum_t (y_t - S^{op}_{1t} - S^{op}_{2t})^2$ under per-period settlement and uniform-across-t allocation in any market that is hourly.

B.2 Equilibrium under hourly schedules ($(1,1)$ and $(1,Q)$ with hourly aggregator)

With both markets hourly the aggregator chooses scalar (S^{op}_1, S^{op}_2) with $S^{op}_1 + S^{op}_2 = \bar{Y}_h$. The per-subperiod implementation is uniform: $S^{op}_{1t} = S^{op}_1/Q$ and $S^{op}_{2t} = S^{op}_2/Q$. Per-subperiod imbalance under per-period settlement is

$$\mathcal{I} \;=\; \sum_t \big(y_t - (S_1^{op} + S_2^{op})/Q\big)^2 \;=\; \sum_t \big((\bar{y}_t - \bar{Y}_h/Q) + \eta_t\big)^2,$$

which depends on $(\bar{y}_t - \bar{Y}_h/Q)$ and η_t, not on the forward-vs-spot split. The aggregator's allocation is therefore driven purely by the price differential: it commits at forward if $p_1 > p_2$ and indifferent if $p_1 = p_2$. We normalise $S_1^{op} = S_2^{op} = \bar{Y}_h/2$ as a tie-break.

B.3 Equilibrium under DA-hourly + ID-granular (state $(1, Q)$)

Notation for forecast resolution. The within-hour heterogeneity that loads onto the aggregator's imbalance has two components: a structural per-subperiod expected production variation $(\bar{y}_t - \bar{Y}_h/Q)$, known already at Stage 1, and a forecast update η_t resolved between Stages 1 and 2 (i.e. between the forward close and the spot close). η_t has variance σ_η^2 and is the empirically relevant "forecast-error shock": bigger σ_η^2 means more within-hour production information that materialises after the forward allocation is fixed.

Optimal per-subperiod spot schedule. The aggregator chooses per-subperiod $\{S_{2t}^{op}\}_t$ at spot using its Stage-2 forecast $\bar{y}_t^{(2)} \equiv \bar{y}_t + \eta_t$ (the updated expected production at the time of the spot close). Setting

$$S_{2t}^{op*} \;=\; \bar{y}_t^{(2)} - \frac{S_1^{op}}{Q} \;=\; \bar{y}_t + \eta_t - \frac{S_1^{op}}{Q}, \tag{6}$$

matches the Stage-2 forecast net of the uniformly-allocated forward commitment. The realised per-subperiod imbalance under (6) reduces to the residual post-Stage-2 noise ε_t (independent of η_t, with small variance), so the expected aggregate imbalance under per-period scheduling is

$$\mathbb{E}[\,\mathcal{I}_{\text{granular}}\,] \;=\; Q \cdot \mathbb{E}[\,|\varepsilon_t|\,]\,.$$

Under the hourly schedule (§B.2) the aggregator must commit at Stage 1 using the Stage-1 forecast $\bar{y}_t^{(1)} = \bar{y}_t$; the Stage-1-to-Stage-2 update η_t then loads onto the imbalance:

$$\mathbb{E}[\,\mathcal{I}_{\text{hourly}}\,] \;=\; \sum_{t=1}^{Q} \mathbb{E}\big[\,\big|(\bar{y}_t - \bar{Y}_h/Q) + \eta_t + \varepsilon_t\big|\,\big]\,.$$

The aggregator's net gain from per-period spot scheduling is

$$G(\sigma_\eta^2) \;\equiv\; \gamma \sum_t \Big[\mathbb{E}\big|(\bar{y}_t - \bar{Y}_h/Q) + \eta_t + \varepsilon_t\big| \;-\; \mathbb{E}\big|\varepsilon_t\big|\Big], \tag{7}$$

strictly increasing in σ_η^2 (the avoided Stage-1-to-Stage-2 forecast shock) and in $\sigma_a^2 \equiv \mathrm{Var}(\bar{y}_t - \bar{Y}_h/Q)$ (the structural within-hour variation). Both components map empirically to within-hour heterogeneity: clock-hours where production swings within the hour carry larger σ_η^2 and larger σ_a^2.

B.4 Quantity shift $\Delta^*(\sigma_\eta^2)$ and its comparative static

The aggregator's marginal trade-off in S_1^{op} is the forward-vs-spot price differential against the marginal change in G. The first-order condition gives an implicit $\Delta^*(\sigma_\eta^2)$:

$$(p_1 - p_2) \;=\; \gamma \cdot \frac{\partial}{\partial S_1^{op}} \Big[\sum_t \mathbb{E}\big|(\bar{y}_t - (\bar{Y}_h - \Delta^*)/Q) + \eta_t\big|\Big]\Big|_{\Delta^*}. \tag{8}$$

The right-hand side is monotone in Δ^* by convexity of the absolute-value expectation in its argument. Differentiating implicitly in σ_η^2,

$$\frac{\partial \Delta^*}{\partial \sigma_\eta^2} \;=\; \frac{-\,\partial^2 G/\partial S_1^{op}\partial\sigma_\eta^2}{\partial^2 G/\partial (S_1^{op})^2} \;>\; 0,$$

since the numerator is positive (a larger forecast-error variance increases the marginal cost of an extra forward-committed MWh, because the realised $|\eta_t|$ contribution to imbalance does not net out across subperiods under per-period settlement) and the denominator is positive by convexity. This establishes (2): the aggregator's quantity shift from forward to spot is strictly increasing in the forecast-error variance.

C Strategic bidder: optimal two-tranche ladder

We derive the strategic bidder's closed-form two-tranche ladder under uniform-price clearing, the FOCs in (p^L, p^H, k^L), the equilibrium spread $\Delta p^* = \Delta_\mu/(2b)$, and a stylisation note that flags the simplification absorbed into the boundary condition.

C.1 Profit under uniform-price clearing

The strategic bidder is the residual monopolist; there is no higher-priced fringe bid above its top tranche. Under uniform-price clearing (as in OMIE day-ahead and intraday auctions), the clearing price is therefore set by the marginal accepted tranche of the strategic bidder's own ladder: p^L when only the low tranche clears, p^H when both clear. For a delivery product at which the residual-demand intercept is $\bar{A} + \mu$ (with $\mu \sim \text{Uniform}[-\Delta_\mu, +\Delta_\mu]$), the strategic bidder commits a ladder $\{(p^L, k^L), (p^H, k^H)\}$ with $p^L \le p^H$, $k^L + k^H = K$ *before* μ realises. The cleared MWh at μ is

$$q(\mu) = \begin{cases} k^L = q^L & \text{if } \mu \le \mu^*, \\ k^L + k^H = K & \text{if } \mu > \mu^*, \end{cases} \qquad \mu^* \;\equiv\; k^L + b\,p^L - \bar{A},$$

where b is the market thinness (either b_{1t} or b_{2t} depending on which market the ladder is on). The clearing price is uniformly p^L when only the low tranche clears and uniformly p^H when both clear — the marginal accepted tranche price, applied to the entire accepted volume. Expected profit per delivery product is therefore

$$\mathbb{E}[\pi] \;=\; \frac{1}{2\Delta_\mu}\left[\int_{-\Delta_\mu}^{\mu^*} (p^L - c)\, q^L\, d\mu \;+\; \int_{\mu^*}^{\Delta_\mu} (p^H - c)\, K\, d\mu\right], \tag{9}$$

with $(p^H - c)\,K$ in the second integrand reflecting that all K units fetch p^H when the high tranche is marginal.

C.2 First-order conditions

Take derivatives in (p^L, p^H, k^L) holding $K = k^L + k^H$ fixed. Define $w^L = k^L/K$, $w^H = k^H/K$.

FOC in p^H. $\frac{\partial \mathbb{E}[\pi]}{\partial p^H} = \frac{1}{2\Delta_\mu}\int_{\mu^*}^{\Delta_\mu} K\, d\mu = \frac{K(\Delta_\mu - \mu^*)}{2\Delta_\mu} > 0$ unless $\mu^* = \Delta_\mu$; at an interior optimum the bidder pushes p^H to the boundary of acceptance, with the binding condition $p^H = (\bar{A} + \Delta_\mu - K)/b$ at the highest μ. Substituting $\mu^* = 0$ at the symmetric solution gives $p^{H*} = (\bar{A} + bc)/(2b) + \Delta_\mu/(4b)$.

FOC in p^L.

$$\frac{\partial \mathbb{E}[\pi]}{\partial p^L} \;=\; \frac{1}{2\Delta_\mu}\left[\int_{-\Delta_\mu}^{\mu^*} q^L\, d\mu \;+\; (p^L - c)q^L \cdot \frac{\partial \mu^*}{\partial p^L}\right] \;=\; \frac{1}{2\Delta_\mu}\big[(\mu^* + \Delta_\mu)\, q^L \;+\; (p^L - c)q^L\, b\big] \;=\; 0.$$

Solving: $\mu^* + \Delta_\mu + b(p^L - c) = 0$, i.e. $\mu^* = -\Delta_\mu - b(p^L - c)$. Combined with the definition $\mu^* = k^L + bp^L - \bar{A}$, this gives $k^L = \bar{A} - bp^L + \mu^*$. At $\mu^* = 0$: $p^{L*} = (\bar{A} + bc)/(2b) - \Delta_\mu/(4b)$.

FOC in k^L.

$$\frac{\partial \mathbb{E}[\pi]}{\partial k^L} \;=\; \frac{p^L - c}{2\Delta_\mu}(\mu^* + \Delta_\mu) \;-\; \frac{p^H - c}{2\Delta_\mu}(\Delta_\mu - \mu^*) \;=\; 0.$$

At the symmetric optimum $\mu^* = 0$ this reduces to $p^L - c = p^H - c$ when $\Delta_\mu - \mu^* = \mu^* + \Delta_\mu$, but the asymmetric prices $p^L = p^{L*}$, $p^H = p^{H*}$ require $\mu^* = 0$ exactly. Substituting $\mu^* = 0$ in $k^L = \bar{A} - bp^{L*}$ and the closed form for p^{L*},

$$k^{L*} \;=\; \bar{A} - b\left(\frac{\bar{A} + bc}{2b} - \frac{\Delta_\mu}{4b}\right) \;=\; \frac{\bar{A} - bc}{2} + \frac{\Delta_\mu}{4}.$$

Closed-form interior optimum.

$$\boxed{p^{L*} = \frac{\bar{A} + b\,c}{2b} - \frac{\Delta_\mu}{4b}, \quad p^{H*} = \frac{\bar{A} + b\,c}{2b} + \frac{\Delta_\mu}{4b}, \quad \mu^{**} = 0, \quad k^{L*} = \frac{\bar{A} - b\,c}{2} + \frac{\Delta_\mu}{4}.} \tag{10}$$

Comparative statics. From (10), the optimal pre-realisation tranche spread is

$$\Delta p^* \;=\; p^{H*} - p^{L*} \;=\; \frac{\Delta_\mu}{2b}. \tag{11}$$

Running the same argument on the hourly aggregate $\tilde{\mu} = Q^{-1} \sum_t \mu_t$ (variance $\sigma_\mu^2/Q < \sigma_\mu^2$) gives a strictly smaller optimal spread, $\Delta p^*|_{\text{hourly}} = \tilde{\Delta}_\mu/(2b)$ with $\tilde{\Delta}_\mu = \Delta_\mu/\sqrt{Q}$. The per-period ladder is therefore strictly richer than its hourly counterpart by a factor $\sqrt{Q}$.

Stylisation note. The closed-form (10) requires that the capacity K be tuned to the boundary condition p^{H*} binds at the highest demand realisation $\mu = \Delta_\mu$; for general K the FOC in p^L also picks up a term $-bK(p^H - c)$ from the second integrand of (9), and the optimum shifts off the symmetric form. The closed form is therefore a stylised approximation, with the auction microfoundation absorbed into the boundary condition. Comparative statics in Δ_μ and b are robust to this stylisation: Δp^* widens with the demand range and contracts with thinness.

Empirical measures of within-curve bid-shape dispersion (MW-weighted price SDs, effective tranche counts) are motivated by this widening but are not derived from the two-state reduction — they require richer bid-curve structure than the two-tranche approximation supports.

D (1, 1) equilibrium

We solve the pre-reform game by backward induction: the Stage 2 best response (§D.1), then the four arbitrage regimes at Stage 1 (§D.2). The scarcity wedge $p_1 > p_2 > c$ survives in regimes (1), (3), and (4); only competitive full arbitrage closes it.

D.1 Stage 2 best response

The strategic bidder's Stage 2 problem, taking p_1, q_1 and s as given:

$$\max_{p_2}(p_2 - c)q_2 \quad \text{s.t.} \quad q_2 = b_2(p_1 - p_2) + s - S_2^{op}.$$

FOC: $b_2(p_1 - p_2) + s - S_2^{op} + (p_2 - c)(-b_2) = 0$, giving

$$p_2^*(p_1, s) \;=\; \frac{p_1 + c}{2} + \frac{s - S_2^{op}}{2\,b_2}. \tag{12}$$

D.2 Stage 1 best response and equilibrium in (p_1, s)

Substituting (12) into the strategic bidder's total profit and using $q_1 = \bar{A} - b_1 p_1 - s - S_1^{op}$,

$$\Pi(p_1, s) \;=\; (p_1 - c)(\bar{A} - b_1 p_1 - s - S_1^{op}) \;+\; (p_2^*(p_1, s) - c)q_2^*(p_1, s).$$

The four arbitrage cases pin s differently.

(R1) No arbitrage ($s = 0$). The strategic bidder solves a Cournot problem in each market separately. The forward FOC gives $p_1^* = (\bar{A} - S_1^{op} + b_1 c)/(2b_1)$. Combining with (12) yields $p_1^* > p_2^* > c$ and $q_1, q_2 > 0$ (Ito and Reguant Result 1).

(R2) Full arbitrage ($p_1 = p_2$). Setting $p_1 = p_2$ in (12) gives $s = b_2(p_1 - c) + S_2^{op}$. Substituting back, total cleared MWh $q_1 + q_2 = \bar{A} - b_1 p_1 - S_1^{op} - S_2^{op} = \bar{A} - b_1 p_1 - \bar{Y}_h$ (using $S_1^{op} + S_2^{op} = \bar{Y}_h$). The strategic bidder maximises $(p_1 - c)(\bar{A} - b_1 p_1 - \bar{Y}_h)$, giving

$$p_1^{*,F} \;=\; \frac{\bar{A} - \bar{Y}_h + b_1 c}{2b_1}, \qquad p_2^{*,F} \;=\; p_1^{*,F}. \tag{13}$$

Note that $s > 0$ *reduces* the strategic bidder's total output relative to no arbitrage (Ito and Reguant Result 2): even competitive arbitrage does not restore competitive prices, because the bidder withholds more in response.

(R3) Limited arbitrage ($s \le K$ binding). If K is below the full-arbitrage level, $s = K$ binds and $p_1 - p_2 > 0$. From (12), $p_1 - p_2 = (p_1 - c)/2 - (K - S_2^{op})/(2b_2)$; the wedge depends on K and on S_2^{op} (Ito and Reguant Result 3). The smaller K, the larger the residual wedge.

(R4) Strategic arbitrage. The arbitrageur maximises $\pi^A(s) = s(p_1 - p_2)$ subject to (12):

$$\pi^A(s) \;=\; s\left[\frac{p_1 - c}{2} - \frac{s - S_2^{op}}{2\,b_2}\right].$$

FOC in s:

$$\frac{p_1 - c}{2} - \frac{2s - S_2^{op}}{2\,b_2} \;=\; 0 \quad\Longleftrightarrow\quad s^{*,S}(p_1) \;=\; \frac{b_2(p_1 - c)}{2} + \frac{S_2^{op}}{2}.$$

Substituting into (12) (with $S_2^{op} \to 0$ for clarity in the wedge derivation),

$$p_2^{*,S} \;=\; \frac{p_1 + c}{2} + \frac{b_2(p_1 - c)/2}{2\,b_2} \;=\; \frac{p_1 + c}{2} + \frac{p_1 - c}{4} \;=\; \frac{3p_1 + c}{4}. \tag{14}$$

The forward-spot wedge is $p_1 - p_2^{*,S} = (p_1 - c)/4$, matching (1) in the main body. (Ito and Reguant Result 4.) The Stage 1 forward price $p_1^{*,S}$ is then the maximiser of the strategic bidder's compound profit; algebra in the standard quadratic gives $p_1^{*,S} - c = 8(\bar{A} - b_1 c - \bar{Y}_h)/(16b_1 - b_2)$ at $S_2^{op} = 0$, $S_1^{op} = \bar{Y}_h$. The wedge $(p_1^{*,S} - c)/4 > 0$ persists at the equilibrium price.

E $(1, Q)$ equilibrium

Forward hourly, spot per-period. We derive the per-product spot best response, work through full and strategic arbitrage to obtain the within-block dispersion equations, and finally compute the within-hour wedge SD in this asymmetric state.

The strategic bidder's Stage 2 problem now has Q spot products. Directly from the spot residual demand $q_{2t} = b_{2t}(p_1 - p_{2t}) + s_t - S_{2t}^{op}$ in (4) and the FOC $q_{2t} = b_{2t}(p_{2t} - c)$, the strategic per-period spot best response is

$$p_{2t}^*(p_1, s_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{s_t - S_{2t}^{op}}{2\, b_{2t}}. \tag{15}$$

The cross-product variation in p_{2t}^* in this state comes purely from per-subperiod variation in b_{2t}, s_t, S_{2t}^{op} — the forward price p_1 is hourly and shared, and μ_t does not enter the spot demand directly (it acts only through the forward-cleared volume, which is uniform per subperiod when forward is hourly).

E.1 Full arbitrage: block parity and within-block dispersion

Block parity requires $p_1 = (1/Q)\sum_t p_{2t}$. Substituting (15),

$$p_1 \;=\; \frac{p_1 + c}{2} + \frac{1}{2Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}} \quad\Longleftrightarrow\quad p_1 - c \;=\; \frac{1}{Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}}.$$

Setting $S_{2t}^{op} \equiv 0$ and uniform arbitrage $s_t \equiv \bar{s}$ to isolate the thinness mechanism, and $Q = 2$ for concreteness,

$$p_1 - c \;=\; \frac{\bar{s}}{2}\left(\frac{1}{b_{21}} + \frac{1}{b_{22}}\right) \;=\; \frac{\bar{s}}{2} \cdot \frac{b_{21} + b_{22}}{b_{21} b_{22}} \quad\Longleftrightarrow\quad \bar{s} \;=\; \frac{2(p_1 - c)\, b_{21} b_{22}}{b_{21} + b_{22}}.$$

The within-block dispersion follows from differencing p_{2t}^* across t:

$$p_{21} - p_{22} \;=\; \left(\frac{\bar{s}}{2b_{21}} - \frac{\bar{s}}{2b_{22}}\right) \;=\; \frac{\bar{s}}{2} \cdot \frac{b_{22} - b_{21}}{b_{21}\, b_{22}}.$$

Substituting $\bar{s}$ from above,

$$\boxed{p_{21} - p_{22} \;=\; (p_1 - c) \cdot \frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(full arbitrage).} \tag{16}$$

E.2 Strategic arbitrage: block wedge and attenuated dispersion

The strategic arbitrageur internalises price impact and stops at

$$p_1 - (1/Q)\sum_t p_{2t} \;=\; \frac{p_1 - c}{4}.$$

The block-wedge condition is the average of (1) across the Q matched spot products. Repeating the algebra of §E.1 with this modified parity, the arbitrage position $\bar{s}$ falls to exactly half of the full-arbitrage value:

$$\bar{s}^S \;=\; \frac{(p_1 - c)\, b_{21} b_{22}}{b_{21} + b_{22}} \;=\; \frac{\bar{s}^F}{2}.$$

The within-block dispersion correspondingly becomes

$$\boxed{p_{21} - p_{22} \;=\; \frac{p_1 - c}{2} \cdot \frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(strategic arbitrage).} \tag{17}$$

The dispersion is attenuated by a factor of two relative to the full-arbitrage case, reflecting the smaller arbitrage volume traded under strategic incentives. The mechanism is dampened but not eliminated.

E.3 Two-tranche ladder on the spot side

The strategic bidder posts per-product spot ladders. Applying §C with $b \to b_{2t}$ gives ladder spread $\Delta p^*_{2t} = \Delta_\mu/(2b_{2t})$, strictly larger than the spread under hourly spot (which uses $\tilde{\Delta}_\mu = \Delta_\mu/\sqrt{Q}$). This is the spot-side richer-ladder mechanism reported in the main body, eq. (3).

E.4 Wedge SD in the asymmetric state

The per-subperiod wedge $w_t = p_1 - p^*_{2t}$ inherits the per-period variation in p^*_{2t}; p_1 is constant across t. From (15), p^*_{2t} varies across t only through per-subperiod thinness b_{2t} (and the arbitrage and aggregator terms when those vary by t). For two matched products ($Q = 2$) the population SD of w_t equals half the within-block price difference, so the full and strategic arbitrage benchmarks give

$$\mathrm{sd}^F_t(w_t) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{2(b_{21} + b_{22})}, \qquad \mathrm{sd}^S_t(w_t) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{4(b_{21} + b_{22})}, \tag{18}$$

the strategic value exactly half of the full-arbitrage value, reflecting the smaller arbitrage volume traded under strategic incentives. Under limited arbitrage with per-product capacity $s_t = K/Q$ binding, the wedge becomes $w_t = (p_1 - c)/2 - K/(2Qb_{2t})$, and

$$\mathrm{sd}^L_t(w_t) \;=\; \frac{K\,|b_{22} - b_{21}|}{8\,b_{21}\,b_{22}} \qquad (Q = 2,\; s_t = K/2), \tag{19}$$

which scales with K rather than with $(p_1 - c)$ but inherits the same $|b_{22} - b_{21}|$ thinness asymmetry. All three benchmarks scale with per-subperiod thinness asymmetry rather than with σ_μ — consistent with the main-body characterisation in §1.4(M3).

F (Q, Q) equilibrium

Both markets per-period. We derive the strategic forward best response per product, work through full, strategic, and (the empirically relevant) limited-arbitrage regimes, and close with the Cournot scale-up of cleared MWh and the Stage-1 option-value mechanism behind the cross-market substitution of bid shapes.

The strategic per-period forward best response from (4) with $\bar{A}_t = \bar{A} + \mu_t$:

$$p^*_{1t} \;=\; \frac{\bar{A} + \mu_t - S^{op}_{1t} + b_{1t}\,c}{2\,b_{1t}}. \tag{20}$$

F.1 Full arbitrage: product parity

Per-product parity $p_{1t} = p_{2t}$ pins s_t at $s_t = b_{2t}(p_{1t} - c) + S^{op}_{2t}$ (same structure as the $(1, 1)$ full-arbitrage condition, but per product). The wedge SD is zero by construction.

F.2 Strategic arbitrage: product wedge and SD

Per-product strategic arbitrage gives (1) applied per product: $p_{2t} = (3p_{1t} + c)/4$. The per-product wedge is $p_{1t} - p_{2t} = (p_{1t} - c)/4$. Cross-period SD:

$$\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \tfrac{1}{4}\,\mathrm{sd}_t(p^*_{1t}).$$

From (20) with S^{op}_{1t} approximately matched to $\bar{y}_t$ (so its variation absorbs structural production variation; the remaining driver is μ_t):

$$\mathrm{sd}_t(p^*_{1t}) \;=\; \mathrm{sd}_t\!\left(\frac{\mu_t}{2b_1}\right) \;=\; \frac{\sigma_\mu}{2\,b_1}.$$

Therefore

$$\boxed{\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \frac{1}{4}\cdot\frac{\sigma_\mu}{2\,b_1} \;=\; \frac{\sigma_\mu}{8\,b_1}.} \tag{21}$$

This is strictly smaller than the asymmetric-state wedge SD $\sigma_\mu/(2b_2)$ whenever $b_2 \le 4b_1$, a condition that holds under the Ito and Reguant assumption $b_2 \le b_1$.

F.3 Two-tranche ladder on the forward side

Applying §C with $b \to b_{1t}$ gives forward-side per-product ladder spread $\Delta p^*_{1t} = \Delta_\mu/(2b_{1t})$, mirroring the spot-side result of §E.3.

F.4 Limited arbitrage at the symmetric state (empirical regime)

The strategic-arbitrage wedge (21) assumes the arbitrageur can build any per-product position. With Q matched products per clock-hour the arbitrageur's capacity K is spread over Q positions, so the effective per-product capacity is $K_t \approx K/Q$. When this capacity binds — which is the empirically relevant case in electricity (see §1.6 "Arbitrage regime is empirical, not theoretical") — the per-product equilibrium is Ito and Reguant Result 3 (limited arbitrage) applied product by product.

Strategic spot best response under binding per-product capacity. Setting $s_t = K_t$ in the strategic Stage 2 FOC (15):

$$p^*_{2t} \;=\; \frac{p^*_{1t} + c}{2} + \frac{K_t - S^{op}_{2t}}{2\,b_{2t}}.$$

The forward per-product cleared price p^*_{1t} depends on μ_t through the forward demand $\bar{A}_t = \bar{A} + \mu_t$; from (20) (corrected to include the arbitrage position), $p^*_{1t} \;=\; (\bar{A} + \mu_t - S^{op}_{1t} - s_t + b_{1t}c)/(2b_{1t})$ so $\mathrm{sd}_t(p^*_{1t}) = \sigma_\mu/(2b_1)$ (under uniform $s_t = K_t$ and S^{op}_{1t} matched to structural production). The per-product wedge is

$$p^*_{1t} - p^*_{2t} \;=\; \frac{p^*_{1t} - c}{2} \;-\; \frac{K_t - S^{op}_{2t}}{2\,b_{2t}}.$$

Wedge SD. Taking the cross-product SD:

$$\mathrm{sd}_t(p^*_{1t} - p^*_{2t}) \;=\; \tfrac{1}{2}\,\mathrm{sd}_t(p^*_{1t}) \;=\; \frac{\sigma_\mu}{4\,b_1}.$$

Compare to the strategic-arbitrage closed-form (21): under unlimited strategic arbitrage the wedge is $(p_{1t} - c)/4$ giving wedge SD $\sigma_\mu/(8\,b_1)$; under limited arbitrage the wedge is $(p_{1t} - c)/2$ giving wedge SD $\sigma_\mu/(4\,b_1)$, exactly twice the strategic-arbitrage value. The wedge SD does *not* collapse fully even at the symmetric state. If per-period thinness b_{2t} also varies across t (as in the asymmetric state) the wedge inherits an additional thinness-asymmetry component on top of $\sigma_\mu/(4b_1)$; the empirically observed sustaining of wedge SD at granular-state magnitudes is consistent with this combined effect.

Per-product capacity binding. For the limited regime to bind, K_t must be smaller than the strategic-arbitrage equilibrium $s_t^{\text{strat}} = b_{2t}(p_{1t} - c)/2$. With $K_t \approx K/Q$ and s_t^{strat} proportional to a per-product mark-up, binding requires $K \leq Q \cdot \bar{s}^{\text{strat}}$ where $\bar{s}^{\text{strat}}$ is the average per-product strategic position. Empirically, hourly arbitrage capacity is set by participation rules and balance-sheet limits that do not scale with Q when granularity reforms multiply products; the constraint binds at the granular state by construction.

F.5 Cournot cleared-MWh under per-period thinness

The strategic best response (20) gives per-period cleared quantity $q_{1t}^* = (\bar{A}_t - b_{1t}c - S_{1t}^{op} - s_t)/2$. Summing across t and taking expectation over μ_t (so $\mathbb{E}[\sum_t \bar{A}_t] = Q\bar{A}$):

$$\mathbb{E}\Big[\sum_t q_{1t}^*\Big] \;=\; \frac{Q\bar{A} - c\sum_t b_{1t} - \sum_t S_{1t}^{op} - \sum_t s_t}{2}.$$

The hourly counterpart is $Q \cdot q_1^*|_{(1,1)} = Q \cdot (\bar{A} - b_1 c - S_1^{op} - s)/2$. Comparing (with the aggregator and arbitrage totals comparable across configurations):

$$\mathbb{E}\Big[\sum_{t=1}^{Q} q_{1t}^*\Big]\Big|_{(Q,Q)} \;-\; Q \cdot q_1^*\big|_{(1,1)} \;=\; \frac{c}{2}\left(Q\, b_1 - \sum_{t=1}^{Q} b_{1t}\right). \tag{22}$$

Positive whenever $\sum_{t=1}^{Q} b_{1t} < Q\, b_1$ — i.e. when per-period thinness is on average smaller than the hourly aggregate elasticity, as expected (each 15-min product is thinner than the hourly aggregate it replaces). This is the Cournot scale-up of main-body §1.5(iv), obtained without any imported mechanism.

F.6 Stage-1 option value (cross-market substitution)

Stage-1 expected profit including Stage 2 is, at second order in μ_t,

$$\mathbb{E}[\Pi(p_1)] \;=\; \pi_1(p_1) + \mathbb{E}_\mu[\pi_2(p_1, \mu_t)] \;=\; \pi_1(p_1) + \pi_2(p_1, 0) + \tfrac{1}{2}\sigma_\mu^2 \cdot \frac{\partial^2 \pi_2}{\partial \mu_t^2}\bigg|_{\mu_t=0} \quad \text{(quadratic-in-}\mu_t\text{; no higher-order rem}$$

Per-period spot Cournot makes π_2 convex in μ_t (since the per-product cleared quantity scales with $\mu_t/(2b_{2t})$ and the per-product mark-up scales with $\mu_t/(2b_{2t})$ via (15), the product is quadratic in μ_t). The second derivative is positive: the variance correction *raises* Stage-1 expected profit. The Stage-1 FOC in p_1 shifts accordingly: the bidder optimises against a more valuable Stage-2 option, widening the forward ladder to capture both sides of the variation. Sign is unambiguous; magnitude depends on $\sigma_\mu^2/(b_{1t}b_{2t})$. Symmetrically when the forward is per-period: the within-hour discrimination is absorbed at Stage 1 directly, so the Stage-2 option value drops and the spot ladder tightens. This is the model-internal account of the cross-market substitution pattern of empirical §4.C.3.

G Balancing discrepancy

We collect the balancing-stage discrepancy under each state, summing demand-side unmet load (μ_t that no auction absorbed) and aggregator production-side unmet schedule (η_t that no per-period spot absorbed). These two channels are what the imbalance penalty γ prices in welfare (§I).

The balancing-stage discrepancy is the difference between realised demand plus aggregator under-/over-production and the auction-scheduled supply, summed per subperiod. With $\bar{A}_t = \bar{A} + \mu_t$ and aggregator realisation $y_t = \bar{y}_t + \eta_t$,

$$\Delta_t \;=\; (\bar{A}_t - q_t^{\text{cleared}}) \;+\; (y_t - S_t^{op}) \;+\; \varepsilon_t, \tag{23}$$

where $S_t^{op} = S_{1t}^{op} + S_{2t}^{op}$ is the aggregator's total scheduled MWh at subperiod t and ε_t is residual noise unrelated to μ_t and η_t.

State $(1,1)$. Forward and spot are both hourly: the forward and spot schedules clear uniformly per subperiod, so per-subperiod demand variation μ_t is unabsorbed; the aggregator commits to a Stage-1 uniform per-subperiod supply and the realisation shock η_t loads onto its schedule deviation. Under the structural assumption $\bar{y}_t \equiv \bar{Y}_h/Q$ (no within-hour structural production variation), $\Delta_t = \mu_t + \eta_t + \varepsilon_t$.

State $(1,Q)$. The spot becomes per-period: the aggregator re-schedules at Stage 2 using its updated forecast $\bar{y}_t + \eta_t$, absorbing the η_t shock into its spot schedule (§B.3). The forward remains hourly so μ_t is still unabsorbed: $\Delta_t = \mu_t + \varepsilon_t$.

State (Q,Q). The forward also becomes per-period: per-product forward clearing q_{1t} responds to $\bar{A}_t = \bar{A} + \mu_t$ through (20), absorbing μ_t. The aggregator's η_t remains absorbed at spot per-period. The balancing discrepancy reduces to $\Delta_t = \varepsilon_t$.

These reductions are mechanical in the contracting grid and hold under any arbitrage regime.

H Note on the alternative sequence

The mirror path $(1,1) \to (Q,1) \to (Q,Q)$ swaps the order of the two auction reforms. After relabelling, the same per-state algebra of §D–F applies. The end-state (Q,Q) is identical. The intermediate state $(Q,1)$ produces a within-hour DA dispersion driven by per-subperiod b_{1t} asymmetry (mirror of (16)–(17) but with roles of b_{1t} and b_{2t} swapped); the aggregator's quantity-shift mechanism activates already at the forward reform; the balancing-discrepancy gain $\Delta_t : \mu_t + \varepsilon_t \to \varepsilon_t$ realises at the forward reform under either path. The peak wedge SD ($\sigma_\mu/(2\,b_2)$ in the original path, $\sigma_\mu/(2\,b_1)$ in the mirror) and the final wedge SD ($\sigma_\mu/(8\,b_1)$ in both) are sequence-invariant up to the relabelling of the dominant elasticity.

I Welfare derivations

We collect the closed-form welfare components by state and arbitrage regime under uniform $\mu_t \sim [-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim [-\Delta_\eta, \Delta_\eta]$ (residual noise ε_t negligible). The aggregator's penalty enters $\pi^{Agg,\text{net}}$ through η_t and the system imbalance cost C^I_{sys} enters through μ_t; the two channels are disjoint.

I.1 Components

Per clock-hour:

$$W \;=\; CS \;+\; \pi^{SB} \;+\; \pi^{Agg,\text{net}} \;+\; \pi^{Arb} \;-\; C^I_{\text{sys}}. \tag{24}$$

The aggregator's net profit $\pi^{Agg,\text{net}}$ is its revenue $\sum_t (p_1^{(t)} S_{1t}^{op} + p_{2t} S_{2t}^{op})$ minus its internalised quadratic penalty $\gamma\, E[\mathcal{I}^{Agg}] = \gamma \sum_t E[(\eta_t^{\text{unhedged}})^2]$ (§B). The system imbalance cost is $C^I_{\text{sys}} = \gamma \sum_t E[\Delta_t^2]$ with Δ_t from §G. $\pi^{Arb} = s(p_1 - p_2)$ is the arbitrageur's revenue ($\pi^{Arb} = 0$ under no arbitrage and full arbitrage). For linear residual demand $q = \bar{A} - bp$, consumer surplus at p^* is

$$CS \;=\; \frac{(\bar{A} - bp^*)^2}{2b}. \tag{25}$$

I.2 Imbalance cost decomposition (closed form, uniform shocks)

Under $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$ and $\eta_t \sim U[-\Delta_\eta, \Delta_\eta]$ independent (and $\varepsilon_t \approx 0$): $E[\mu_t^2] = \Delta_\mu^2/3$, $E[\eta_t^2] = \Delta_\eta^2/3$. The aggregator's unhedged shock (§B.2) is η_t when spot is hourly and 0 once spot is per-period; the system discrepancy (§G) is μ_t when forward is hourly and 0 once forward is per-period. The quadratic-penalty costs decompose as

$$
\begin{array}{lccc}
 & (1,1) & (1,Q) & (Q,Q) \\
\hline
\gamma\, E[\mathcal{I}^{Agg}] \text{ (in } \pi^{Agg,\text{net}}) & \gamma Q\Delta_\eta^2/3 & 0 & 0 \\
C^I_{\text{sys}} \text{ (separate)} & \gamma Q\Delta_\mu^2/3 & \gamma Q\Delta_\mu^2/3 & 0 \\
\text{Total imbalance cost} & \gamma Q(\Delta_\mu^2+\Delta_\eta^2)/3 & \gamma Q\Delta_\mu^2/3 & 0
\end{array}
\tag{26}
$$

The two channels are disjoint (η_t only enters $\pi^{Agg,\text{net}}$, μ_t only enters C^I_{sys}). The two reform steps save $\gamma Q\Delta_\eta^2/3$ (spot, absorbed into $\Delta\pi^{Agg,\text{net}}$) and $\gamma Q\Delta_\mu^2/3$ (forward, absorbed into $-\Delta C^I_{\text{sys}}$).

I.3 Welfare at state $(1,1)$ by arbitrage regime

No arbitrage (R1). From §D.2: $p_1^{NA} = (\bar{A}+b_1c-S_1^{op})/(2b_1)$, $q_1^{NA} = (\bar{A}-b_1c-S_1^{op})/2$, $\pi_1^{SB,NA} = (\bar{A}-b_1c-S_1^{op})^2/(4b_1)$. Consumer surplus at the forward stage is, by (25), $CS_1^{NA} = (\bar{A}-b_1p_1^{NA})^2/(2b_1) = (q_1^{NA}+S_1^{op})^2/(2b_1)$. The spot stage adds analogous terms via the spot best response (12). Total welfare:

$$W^{NA}_{(1,1)} \;=\; CS_1^{NA} + CS_2^{NA} + \pi_1^{SB,NA} + \pi_2^{SB,NA} + \pi^{Agg,\text{net},NA} - \gamma Q\Delta_\mu^2/3,$$

with $\pi^{Agg,\text{net},NA}$ including the $-\gamma Q\Delta_\eta^2/3$ internalised penalty.

Full arbitrage (R2). At $p_1 = p_2 = p_1^F = (\bar{A}-\bar{Y}_h+b_1c)/(2b_1)$ from (13), strategic-bidder profit is $\pi^{SB,F} = (p_1^F - c)(\bar{A}-b_1p_1^F-\bar{Y}_h) = (\bar{A}-b_1c-\bar{Y}_h)^2/(4b_1)$. The cleared MWh stays below the competitive benchmark (Ito and Reguant Result 2): with monopoly market power, even competitive arbitrage cannot restore the first-best.

Strategic arbitrage (R4). The arbitrageur extracts $\pi^{Arb,S} = s^{*,S}(p_1-p_2)|_{s=s^{*,S}} = b_2(p_1-c)^2/4$ (§D.2), a pure transfer that subtracts from the strategic bidder's rent. Welfare lies strictly between W^{NA} and W^F.

Limited arbitrage (R3, empirical). With $s = K < b_2(p_1-c)/2$ binding, the arbitrageur extracts $\pi^{Arb,L} = K\,[(p_1-c)/2 - (K-S_2^{op})/(2b_2)]$. Welfare is closer to W^{NA} when K is small, and converges to W^S as K grows to $b_2(p_1-c)/2$.

I.4 Welfare at $(1,Q)$ and (Q,Q) — delta from $(1,1)$

The reform's per-step welfare change decomposes into closed-form imbalance, ladder-rent, and (at (Q,Q)) Cournot scale-up terms.

Spot reform, $(1,1)\to(1,Q)$: aggregator hedging. The aggregator re-schedules per-subperiod at spot (6); $E[\mathcal{I}^{Agg}]$ drops from $Q\,\Delta_\eta^2/3$ to 0, so $\Delta\pi^{Agg,\text{net}} = +\gamma Q\,\Delta_\eta^2/3$.

Spot-side ladder rent. The strategic bidder's spot-side two-tranche ladder (3) replaces a single-price Cournot with a Δ_μ-discriminated ladder. By §C closed form (substituting $b = b_{2t}$):

$$\Delta\pi^{SB,2T}_{\text{spot}} \;=\; \frac{\Delta_\mu\left[4(\bar{A}-b_{2t}c)+\Delta_\mu\right]}{16\,b_{2t}}. \tag{27}$$

On the inframarginal volume $k^{L*} = (\bar{A} - b_{2t}c)/2 + \Delta_\mu/4$, this is a CS-to-$\pi^{SB}$ transfer; on the marginal extension of $\Delta_\mu/2$ MWh, this is new surplus. The net welfare gain is $\Delta_\mu(\bar{A} - b_{2t}c)/(4b_{2t}) + \Delta_\mu^2/(16b_{2t})$ from the extension — a fraction of (27).

Forward reform, $(1, Q) \to (Q, Q)$: system imbalance. Per-product forward absorbs μ_t via (20), so C^I_{sys} drops from $\gamma Q\,\Delta_\mu^2/3$ to 0:

$$\Delta(-C^I_{\text{sys}}) \;=\; +\gamma Q\,\Delta_\mu^2/3.$$

Forward-side ladder rent. The same closed form as (27) applies on the forward side with $b = b_1$:

$$\Delta\pi^{SB,2T}_{\text{forward}} \;=\; \frac{\Delta_\mu\left[4(\bar{A} - b_1 c) + \Delta_\mu\right]}{16\, b_1}. \tag{28}$$

Cournot scale-up at (Q, Q). Total cleared MWh per clock-hour rises by $(c/2)(Qb_1 - \sum_t b_{1t})$ via (22). At the mark-up $p_1 - c = (\bar{A} - b_1 c - S_1^{op})/(2b_1)$,

$$\Delta W^{\text{scale}} \;=\; \frac{c\,(\bar{A} - b_1 c - S_1^{op})}{4\, b_1}\Big(Qb_1 - \sum_t b_{1t}\Big). \tag{29}$$

Limited-arbitrage gain at $(1, Q)$ and (Q, Q). Per-product capacity K/Q binds, leaving a per-product wedge $(p_{1t} - c)/2$ instead of the unlimited-strategic $(p_{1t} - c)/4$ (§F.4). The strategic bidder's spot markup over marginal cost *falls* from $3(p_1 - c)/4$ to $(p_1 - c)/2$, so the per-product deadweight loss falls by $5b_2(p_1 - c)^2/32$ (Ito and Reguant (2016) Result 2: less arbitrage means less strategic withholding). This is the welfare counterpart of the empirical asymmetric-persistence wedge-SD finding — not a welfare cost.

I.5 Net welfare deltas (closed form)

Combining the channels for each reform step (writing $a_m \equiv \bar{A} - b_m c$ for $m \in \{1, 2\}$):

$$\Delta W^{\text{spot}} \;=\; \frac{\gamma Q \Delta_\eta^2}{3} \;+\; \frac{5\, b_2(p_1 - c)^2}{32} \;+\; \frac{\Delta_\mu\,[4a_2 + \Delta_\mu]}{16\, b_2}, \tag{30}$$

$$\Delta W^{\text{forward}} \;=\; \frac{\gamma Q \Delta_\mu^2}{3} \;+\; \frac{c\, a_1}{4\, b_1}\Big(Qb_1 - \sum_t b_{1t}\Big) \;+\; \frac{\Delta_\mu\,[4a_1 + \Delta_\mu]}{16\, b_1}, \tag{31}$$

where the second term of (30) is the per-product limited-arbitrage gain at the regime change, and the third terms in both lines are the bid-ladder rent extensions (mostly distributional).

Sign and dominance. Both deltas are unconditionally positive — the model predicts welfare improvement at each reform step, with imbalance saving the dominant channel given that balancing prices in Spanish electricity markets exceed day-ahead clearing prices by a factor of 2–10.